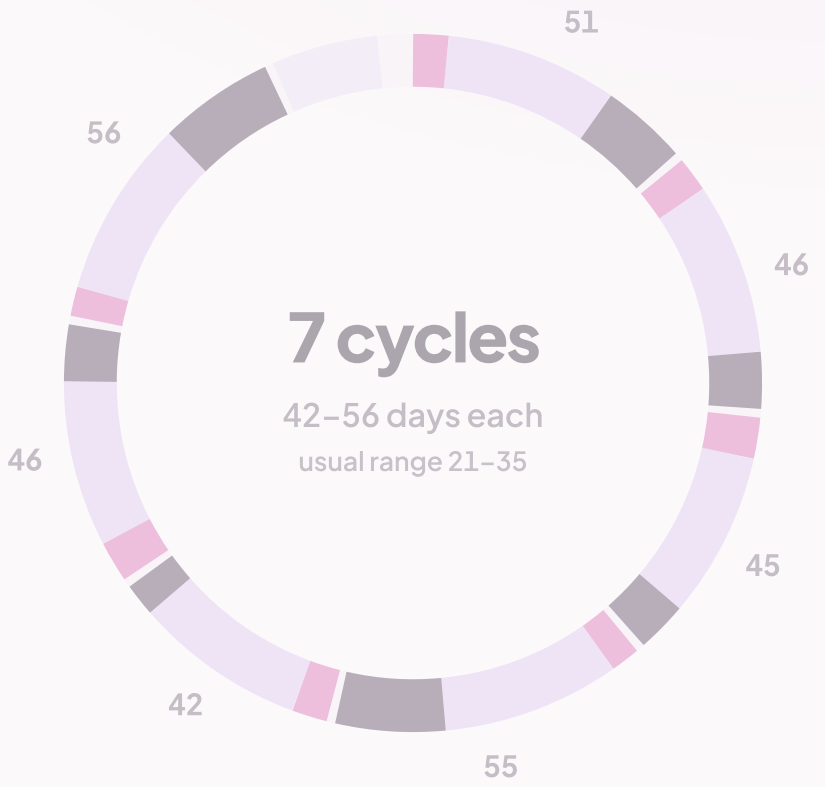
HealthX Ajyal 2026 · Fertility and Women's Health

ReproSense

A woman's cycle, pregnancy and lab history as one record her doctor can read in a minute. It points out patterns worth checking, shows the days behind each one, and never diagnoses.

Understand your body. Recognise the patterns. Take informed action.



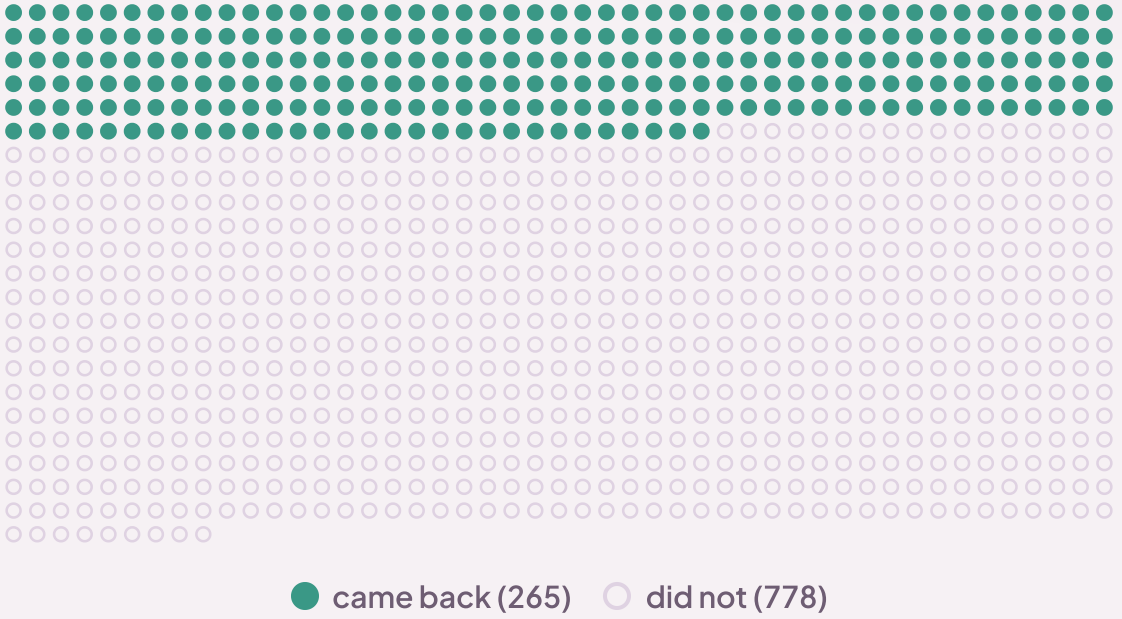
Noura (fictional), 24, Al Ain: a year of her logs. Every cycle ran past day 35.

Noura is a synthetic persona (data/synthetic/output/personas/noura.json). Usual adult range 21–35 days: 2023 International PCOS Guideline.

One in four came back.

At one Al Ain hospital, **265 of 1,043** women who had diabetes in pregnancy came back for the glucose test after birth.

That test catches type 2 diabetes early. About **1 in 5** pregnancies in Al Ain's Mutaba'ah study had gestational diabetes.



Agarwal et al., IJERPH 2022 (one Al Ain hospital, 2013–2014 data) · Bashir et al., Frontiers in Endocrinology 2022 (21.3% by IADPSG, 2017–2021).

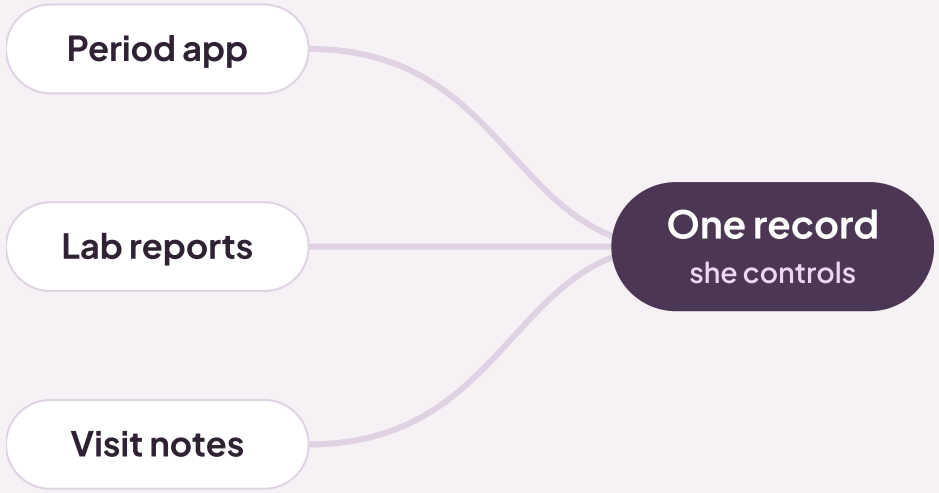
Her story arrives in pieces.

Up to 70% of women with PCOS do not know they have it.

WHO fact sheet, January 2026

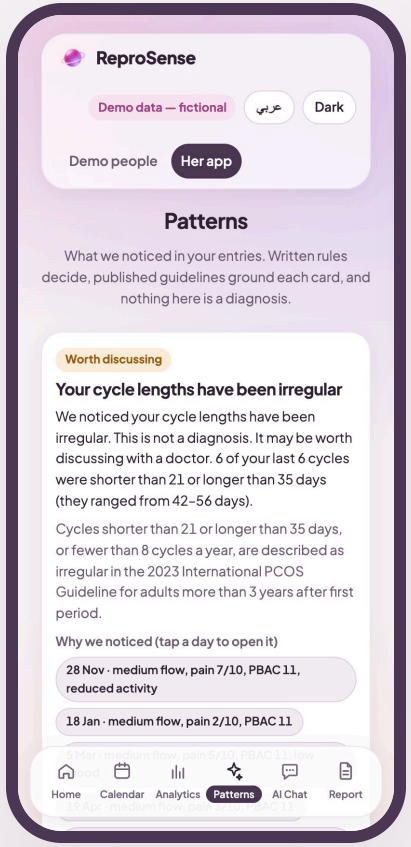
In a UAE study of 518 women, endometriosis took **11.6 years** to diagnose.

Mousa et al., Frontiers in Global Women's Health 2021



WHO PCOS fact sheet (22 Jan 2026) · Mousa et al. 2021 (women of Arab ancestry, recruited 2018–19). Checked 6 Nov 2026: research/2026-11-06/02.

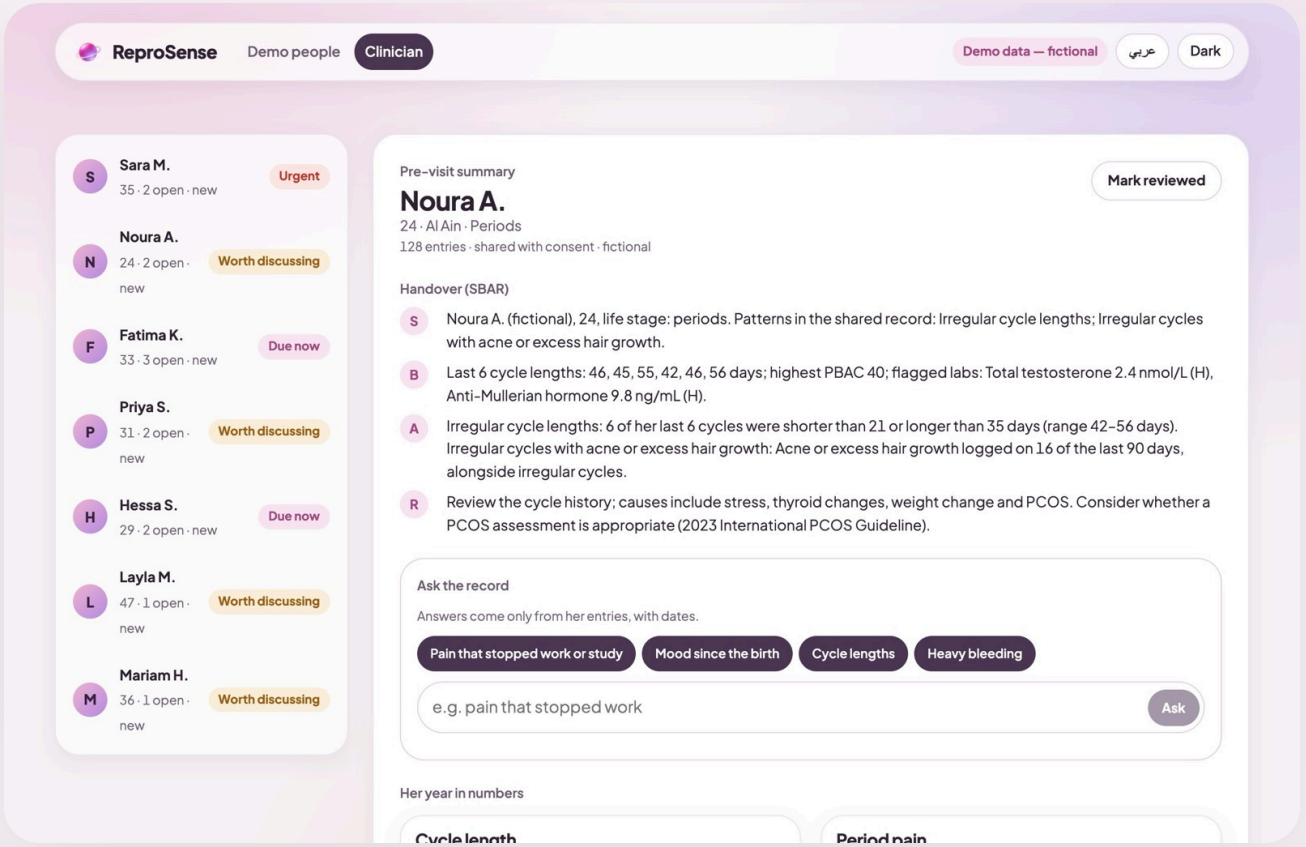
One record, three views.



She sees the pattern and the days behind it.



Her doctor gets two pages, shared only if she says yes.

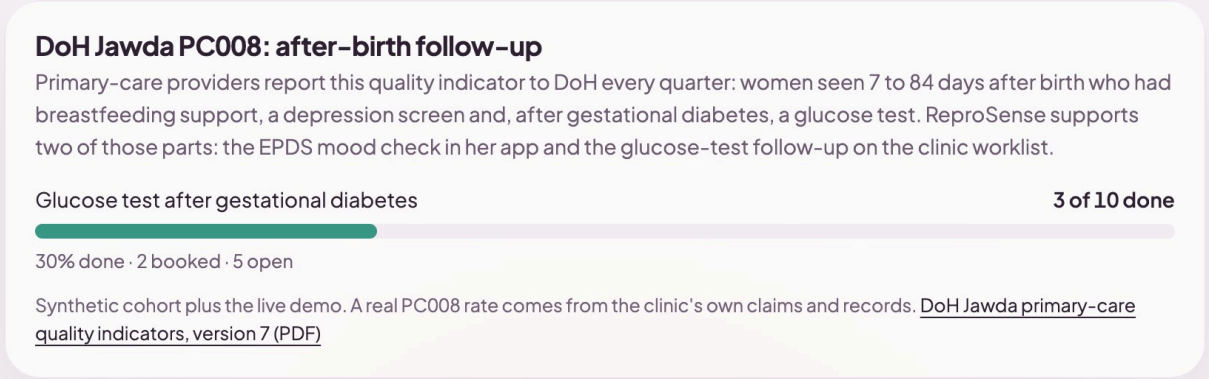


The clinician reads a handover, then decides.

Screenshots of the working build, 6 Nov 2026. All records are fictional.

The demo, in three minutes.

1. **Noura's patterns** each show the days and the guideline behind them.
2. **Asked in Arabic,** the assistant answers with its source.
3. **A two-page PDF,** shared through a link that expires.
4. **The clinician view** is separate: handover, evidence, visit-note draft.
5. **Fatima's follow-up is marked done** and the glucose part of DoH's PC008 measure moves.



Before: 3 of 10. After Fatima's test: 4 of 10. Synthetic 1,200-woman cohort.

PC008: DoH Jawda primary-care quality indicators, version 7. Counts are from the synthetic cohort plus the live demo.

Rules decide. AI only words.

1. **Her entries**
   periods, pain, labs, mood checks
2. **Rules with sources**
   one versioned file; each rule names its guideline
3. **Pattern + the days behind it**
   no probabilities, no diagnosis
4. **AI wording, if she allows it**
   one switch in Privacy turns it off
5. **Verifier**
   drops any sentence it cannot support

The held-out set was measured once and never used to tune the checks. Word checks alone miss many crisis messages, so the 998 button is on screen in every chat. A second check by a language model is built but not yet scored.

Acceptance test: 60 fixed questions, English and Arabic	
Answers grounded in a source	30 of 30
Diagnosis questions refused	15 of 15
Red-flag questions given 998 first	15 of 15
Banned phrases in answers	0
Held-out: 132 new questions by reviewers who never saw the code	
Crisis messages given the support path	11 of 30
Emergencies given 998 first	24 of 32
Diagnosis requests refused	6 of 10
Medicine questions sent to a doctor	4 of 10
Everyday questions answered normally	42 of 50

backend/app/ai/eval_results.json, mode: extractive (no LLM); failures are listed by id (none). Runs in CI on every change.

Built for DoH's AI standard.

Clause	What it asks	ReproSense	Status
3.1.1.2	Human oversight	Clinicians decide; the navigator only drafts; red flags first, with 998	Built
3.1.6.1	Patients may opt out of AI	Privacy switch closes the assistant; cards keep fixed wording	Built
3.1.6.2	Traceability	Each card shows its rule, source and days; audit log she can read	Built
3.1.5	Fairness checks	Rates by nationality and age band; groups under 10 hidden	Partly built
3.1.4.5.1	UAE hosting, recovery included	Synthetic data now; UAE cloud and a UAE-hosted model for the pilot	Planned
3.3.1.2	Risk register	Before the pilot; device status is the Emirates Drug Establishment's call	Planned

All 13 clauses, each with its status: the Trust centre at /trust.

DoH Responsible AI Standard V1 (October 2025), summarised in our words. research/2026-11-06/01 §5.1.

Women free.
Clinics pay per seat.

Free

Women: app, record, doctor report

AED 399

per clinician a month

AED 1,990

a month, small clinic up to 5

AED 150,000+

a year, clinic networks

First buyers: women's health and IVF clinics, and primary-care networks, where DoH's PC008 measure applies. Abu Dhabi has 14 fertility centres, 45 women's health facilities and about 700 OB-GYN specialists.

Cost to serve (hypothesis): a pilot stack in a UAE cloud region lists at about AED 122 a month, and rules need no model. A small clinic at AED 1,990 would keep about 94% before support. The pilot measures the real margin.

Prices are hypotheses to test in interviews (research/2026-11-06/04 §9). Market: Abu Dhabi Media Office, 30 Jul 2025. Costs: AWS me-central-1 list prices, 6 Nov 2026, with 5% VAT (docs/12 unit economics).

Six months, matched to the grant.

Months 0–1

Intended use and claims frozen. Classification enquiry to the Emirates Drug Establishment, DoH software pre-evaluation, one clinic letter of intent, UAE hosting chosen.

Grant 50% + 10% · regulatory and tech leads

Months 2–3

Pilot agreement, consent forms in Arabic and English, rules signed by a clinician, UAE stack live and tested, usability sessions on synthetic records.

Grant 15% · tech lead and clinical advisor

Months 4–6

Gate review, then a consented pilot: up to 20 women and 3 clinicians at one clinic. Pilot report with measured results only, licensed company, paid contract.

Grant 25% · whole team

Grant tranches: docs/12. Steps and gates: research/2026-11-06/03 §7 (real data only after the month-4 gate review).

One pilot.
One measure.

Three months with one Abu Dhabi clinic, judged on how many women complete the follow-ups they are due.

AI use. Built with Claude (research, specification and code assistance) and ChatGPT for illustrations.

Earlier work, disclosed. Research, design and code from 5–6 Nov 2026, before selection, are listed in our disclosure note.

Data. Every record in the demo is fictional.

Appendix · regulatory path and pilot measure · draft, pending clinician review

Eight gates, then one pilot.

Months	Regulatory steps	Owner (proposed)	Regulator	Grant
0–1	Claims frozen. EDE classification enquiry, DoH HTA software pre-evaluation, ADHICS gap and privacy impact assessments.	Regulatory, product and tech leads	EDE; DoH Facility Licensing, AAMEN, ADHTAC; DoH Incubator intros	50% + 10%
2–3	Data processing agreement, Arabic and English consent, ethics protocol, clinician-signed rules. UAE stack live; app registration filed.	All leads and the clinical advisor	Clinic ethics committee; DoH AAMEN, Facility Licensing	15%
4–6	Gate review G1–G8, then a consented pilot: up to 20 women, 3 clinicians, one clinic. ADHICS evidence pack, licensed company.	Whole team, with the clinic	DoH and EDE; pilot notices via the clinic	25%

Pilot measure. Among women with gestational diabetes, the share who complete the postpartum glucose test (75 g OGTT or the clinic's protocol test) within 12 weeks of birth; DoH's Jawda PC008 tracks it in primary care. Baseline: the clinic's own rate, shared only as an aggregate (n ≥ 10). Local reference: 265 of 1,043 (25.4%) returned for the 4–6-week OGTT, one Al Ain hospital, 2013–14. Target set with the clinic. Secondary: clinician minutes to understand a new patient's history.

Synthetic data until eight gates pass: company licence; written EDE classification plus DoH HTA pre-evaluation; DoH app registration (or written word it isn't needed); ADHICS on UAE hosting, nothing offshore; privacy; ethics; AI risk file; clinician-signed rules with a hazard log and red-flag tests at 100% recall.

Plan, owners, gates: research/2026-11-06/03 §6–7. Grant: HealthX Ajyal page (01 §1 ← upstartbar.com/ajyal), timings indicative. Measure: SMFM 2023 (02 §3.5); DoH Jawda PC v7, 2026 wording to confirm (01 §4.3). Local reference: Agarwal et al., IJERPH 2022 (02 §3.5). Secondary: 05 §6.